

NUMERICAL ANALYSIS PROBLEM SET 2

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We begin by noting that all of the files referenced in this document are available at:

<https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet2>

1. Use the command `magic(n)`, for $n = 100, 200, 300, \dots, 10^4$ to construct the magic square matrices of order n . Magic squares are defined as those matrices having entries for which the sum of the elements by rows, columns, or diagonals are identical. Then compute their determinants by the command `det()`. Use the command `cputime` to get the CPU time that is needed for this computation. **Be patient, it may take around 10 minutes to compute these determinants.**